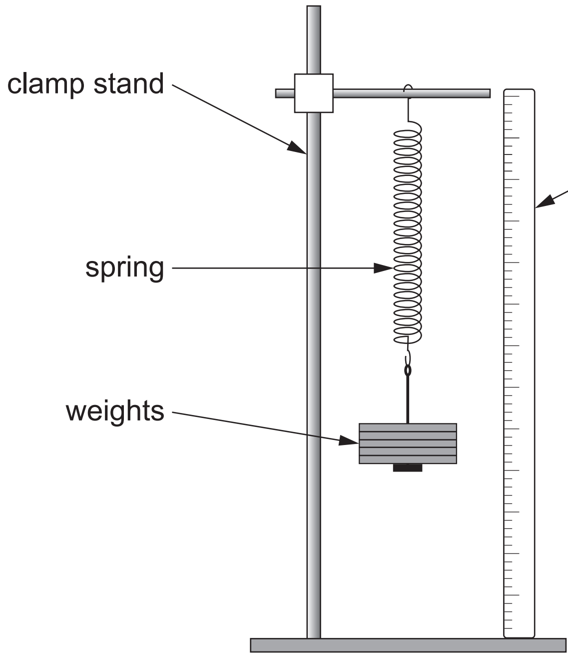
5. The diagram shows the apparatus used by a group of students when investigating the stretching of a spring.



The spring is loaded and then unloaded.
Its extension is determined when different weights are added.
Their results are shown below.

Force (N)	Extension when loading (cm)	Extension when unloading (cm)	Mean extension (cm)
0.0	0.0	0.0	0.0
1.0	3.9	4.1	4.0
2.0	8.0	8.0	8.0
4.0	16.0	16.0	16.0
5.0	20.2	19.8	20.0
6.0	23.7	24.3	24.0
7.0	28.0	28.0	28.0

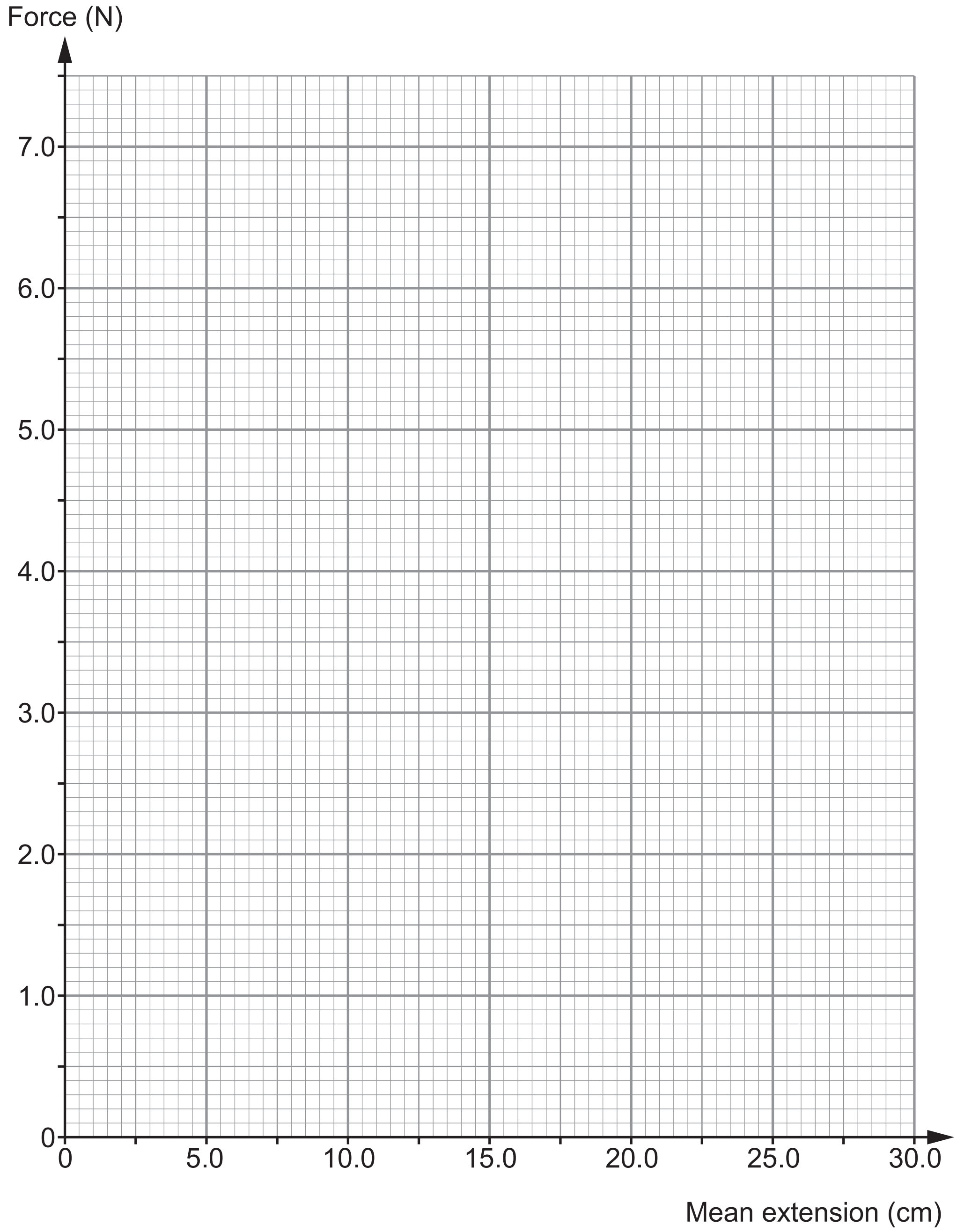
(a) State the **two** measurements that need to be taken to determine the extension of the spring. [2]

1. ....

2. ....

Examiner only

(b) Plot the data on the grid below and draw a suitable straight line. [3]



(c) Use your graph to determine the extension for a force of 2.5 N. [1]

extension = ... cm

Examiner only

(d) Use your answer from part (c) and the equation:

$$\text{spring constant} = \frac{\text{force}}{\text{extension}}$$

to calculate the spring constant for a force of 2.5 N. [2]

spring constant = ... N/cm

(e) Explain how you could check whether the results of this investigation are reproducible. [2]

....

....

....

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